Inorder Successor in BST (medium)

Problem Statement

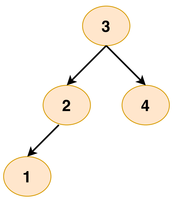
Given a `root` node of the binary search tree and node `p`, return the value of the in-order successor of node `p` in the given tree. If the given node has no in-order successor, return `-1`.

The `in-order` successor of a node is the next node in the in-order traversal of the BST, which means it is the node with the `smallest` value `greater` than the given node.

Examples

Example 1:

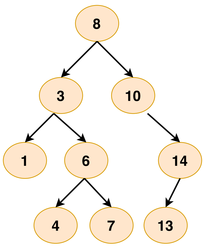
- **Input**: root = [3,2,4,1], p = 2



- **Expected Output**: 3
- **Justification**: In the in-order traversal [1,2,3,4], the next node after 2 is **3**, making it the in-order successor.

Example 2:

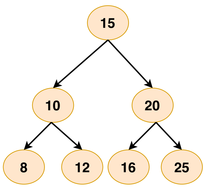
- **Input**: root = [8,3,10,1,6,null,14,null,null,4,7,13], p = 6



- **Expected Output**: 10
- **Justification**: The in-order traversal of the tree is [1,3,4,6,7,8,10,13,14]. The smallest value greater than 6 is **7**, making it the in-order successor.

Example 3:

- **Input**: root = [15,10,20,8,12,16,25], p = 25



- **Expected Output**: -1
- **Justification**: In the in-order traversal [8,10,12,15,16,20,25], there is no node after 25. Hence, the expected output is **-1**.

Try it yourself

Try solving this question here:

Java

```java
// class TreeNode {
//     int val;
//     TreeNode left, right;
//     TreeNode(int x) { val = x; }
// }

public class Solution {
    public int inorderSuccessor(TreeNode root, int p) {
        // ToDO: Write Your Code Here.
        return  -1;
    }
}
```

Test Case | Run Code Result | Stdout

```
[3,2,4,1]
2
```

Use Example Testcases | Run | Submit